Total Printed Pages: 03

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MAB-101
Examination May - 2024
B.Tech. I Sem : Common to all Branches
Linear Algebra and Calculus

Time :2Hrs

Max. Marks :60
Min. Marks :19

Note: Total number of questions are 05. All Questions are compulsory. Each Question has 4 parts (a, b, c, d). Part a & b are compulsory while Part c &d has internal Choice. Assume missing data, if any.
Word limit be observed as follows:
Part a & b – Objective type/Fill in the blanks/One sentence answer.
Part c – Max 100 words and Part d – Max 400 words.
Word limit NOT to be followed for diagram, numerical, derivation.

			Marks
Q.1	(a)	Define the Taylor's theorem.	02
	(b)	The function $x^2 + y^2 - 6x + 10$ is minimum at the point (i) (3, 0) (ii) (3, 3) (iii) (-3, 0) (iv) (-3, -3)	02
	(c)	Expand the $e^{\sin x}$ by Maclaurin's theorem up to the terms containing x^3.	03
		OR	
		Discus the maximum and minimum value of $x^2 + y^2 + 6x + 12$.	03
	(d)	Show that the radius of curvature of the hypocycloid $x^{2/3} + y^{2/3} = a^{2/3}$ at any point (b, c) is$(a \cdot b)^{1/3}$.	05
		OR	
		Discus the maxima and minima of the function $x^3 + y^3 - 3axy$.	05
Q.2	(a)	Define the Homogeneous function.	02
	(b)	State the Euler's theorem.	02
	(c)	If $u = \log(x^3 + y^3 + z^3 - 3xyz)$ show that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = \frac{3}{x+y+z}$	03
		OR	
		If $u = sin^{-1} \frac{x^2+y^2}{x+y}$ show that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \tan u$.	03

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	(d)	If u is a homogeneous function of degree n then $$x^2\frac{\partial^2 u}{\partial x^2}+2xy\frac{\partial^2 u}{\partial x\partial y}+y^2\frac{\partial^2 u}{\partial y^2}=n(n-1)u$$	05
		OR	
		Evaluate $\sqrt{99}$ approximately.	05
Q.3	(a)	Choose the correct alternative $$\lim\sum_{r=1}^{n-1}\frac{1}{n}\sqrt{\frac{n+r}{n-r}}=$$ (i) $\int_0^1\frac{1-x}{\sqrt{1-x^2}}dx$ (ii) $\int_0^n\frac{1-x}{\sqrt{1-x^2}}dx$ (iii) $\int_0^1\frac{1+x}{\sqrt{1-x^2}}dx$ (iv) $\int_0^1\frac{1-x}{\sqrt{1+x}}dx$	02
	(b)	The value of $\int_0^1\int_0^x\int_0^{x+y}dxdydz$ is (i) 1 (ii) $^1/_3$ (iii) $^1/_4$ (iv) $^1/_2$	02
	(c)	Find the limit of the series $$\lim_{n\to\infty}\sum_{r=1}^{n}\frac{r^3}{r^4+n^4}$$	03
		OR	
		Evaluate $$\int_0^1\int_0^{\sqrt{1+x^2}}\frac{dydx}{1+x^2+y^2}$$	03
	(d)	Find the area between the curves $x^2+y^2=a^2$ as a double integral and evaluate it.	05
		OR	
		Evaluate $\int\int\int_R(x+y+z)dxdydz$ R: $0\le x\le 1,\ 1\le y\le 2,\ 2\le z\le 3$	05
Q.4	(a)	Define the adjoint of a matrix.	02
	(b)	Define triangular matrices.	02

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	(c)	If $A = \begin{vmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{vmatrix}$ show that $A^2 - 4A - 5I = 0$.	03
		OR	
		If $A = \begin{vmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{vmatrix}$ find A^{-1}	03
	(d)	Test for consistency and solve $5x + 3y + 7z = 4, \quad 3x + 26y + 2z = 9, \quad 7x + 2y + 10z = 5$	05
		OR	
		Find the Eigen value of the matrix $A = \begin{vmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{vmatrix}$	05
Q.5	(a)	Define the following: (i) Null Graph (ii) Complete Graph (iii) Degree of a vertex (iv) Euler Graph	02
	(b)	Define the connected and disconnected graphs.	02
	(c)	Find the truth tables of the following: (i) $\sim(\sim p \vee \sim q)$ (ii) $\sim(\sim p \wedge q)$	03
		OR	
		Show that the number of odd degree vertices in a graph are always even.	03
	(d)	Define Boolean Algebra and the principal of Duality.	05
		OR	
		(a) Write the Dual statement of each of the following: (i) $(1 + x) \cdot (0 + y) = y$ (ii) $x \cdot (x' + y) = x \cdot y$ (b) Solve the following Boolean function $(x + y) \cdot (\bar{x} \cdot z) \cdot (y + z)$	05

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